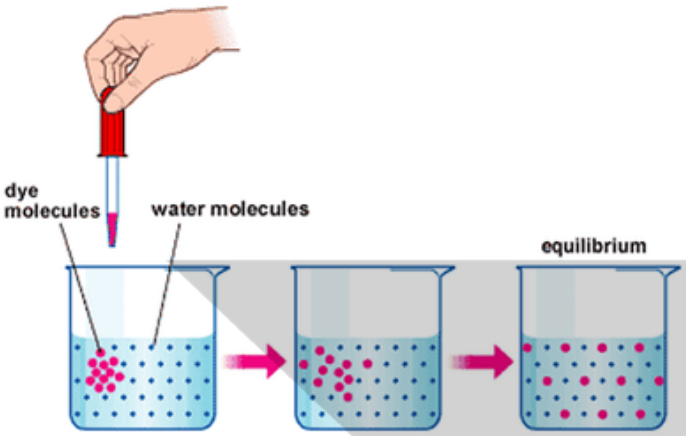
1. Movement Into and Out of Cells

1.1. Diffusion

Diffusion: Net movement of particles down the concentration gradient (high → low) as a result of their **random movement**.



- At **Equilibrium,** the concentrations of molecules are equal.
- Energy for diffusion comes from the **kinetic energy of the random movement of molecules and ions.**
- The diffusion of gases and solutes is important as without it, molecules that are needed for life, (Such as glucose and oxygen for respiration) would not be able to get to the places they are required.
- Some substances move into and out of cells by diffusion through the cell membrane.

Factors that influence diffusion (Passive Transport)

- Concentration gradient
- Temperature
- Surface area to volume ratio
- Distance

1.2. Osmosis

- The role of water acts as **a solvent** in organisms to aid with **digestion, excretion, and transport.**
- Knowing the definition of **Diffusion, Osmosis and Active Transport is COMPULSORY**!
- The cell membrane is partially permeable, allowing small molecules (like water) through but not larger molecules.

Osmosis: Net movement of water molecules from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) through a **partially permeable membrane.**

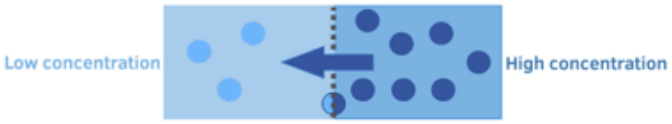


Fig 1. Osmosis. Water moves from and area of high to low concentration.

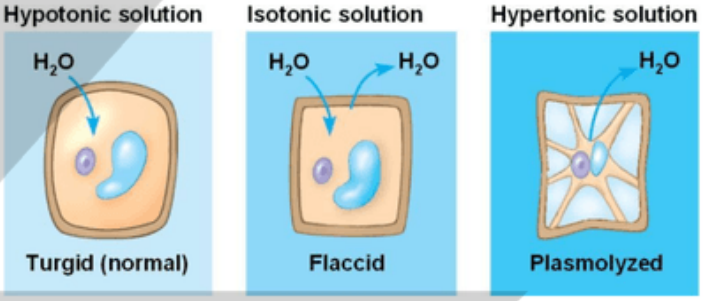
Conc. of Solute (In-Cell)	Conc. of Solute (Outside-Cell)	Condition of the Cell
Low	High	Cell Shrinks (**Flaccid**/Hypertonic)
Same	Same	No Change (Isotonic)
High	Low	**Turgid**/Hypotonic

In Animals

- Increasing solute concentration inside a cell can cause it to burst (**cell lysis**) because it has too much water and no cell wall.

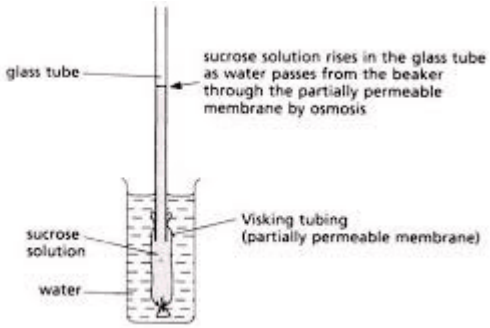
In Plants

- Increasing solute concentration inside the cell causes the cell to become **turgid**, and the vacuole fills up. The cell wall can withstand the **higher turgor pressure**.
- Decreasing solute concentration inside the cell causes the cell to become **flaccid**, losing water, and the vacuole to get smaller. The cell body shrinks, pulling away from the cell wall.
- Plants are supported by the water pressure inside the cells pressing outwards on the cell wall.
- Plasmolysis: the process in which a plant cell loses water due to osmosis, causing the cell membrane to shrink away from the cell wall.



Dialysis Tubing Experiment

- **Dialysis Tubing** (or Visking tubing) is a non-living, **partially permeable** membrane made from cellulose.
- Pores are small enough to **prevent the passage of large molecules** (such as **sucrose**) but allow **smaller molecules** (such as **glucose and water**) to pass through by **diffusion** and **osmosis.**



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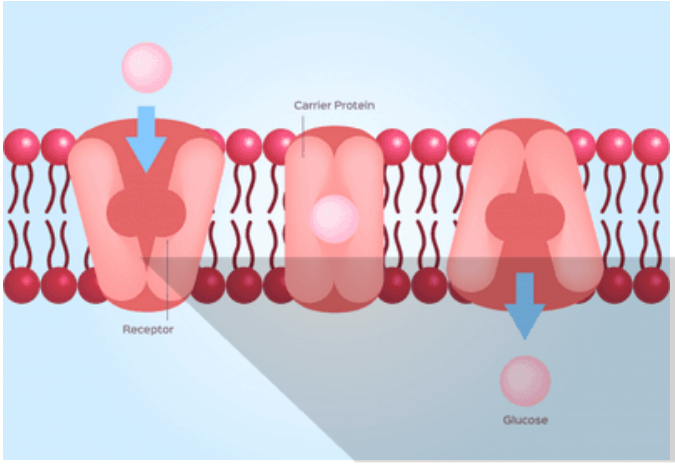
1.4. Active Transport

Active Transport: The movement of particles through a cell membrane from a lower concentration region to a higher concentration region (i.e., against a concentration gradient) using energy from respiration (ATP)

- Carrier Proteins or Protein Carriers are also used during active transport.



- It is embedded in the cell membrane, where it picks up specific molecules and moves them through the membrane against their concentration gradient.
- Active transport is needed when an organism wants to optimise the nutrients it can take up - ion uptake by root hair cells.
- Some particles are too large to cross a membrane by diffusion or active transport. A few very specialised cells have developed a method for taking up these particles; the particles are engulfed by the cell surface membrane flowing around them which is called Endocytosis. Endocytosis (called phagocytosis) is used by white blood cells to engulf pathogens.

1.5. Difference between Diffusion, Osmosis, and Active Transport

Feature	Diffusion	Osmosis	Active Transport
Type of movement	Passive (no energy required)	Passive (no energy needed)	Active (requires energy-ATP)
Direction	High → Low concentration	High → Low water potential	Low → High (against gradient)
Substance moved	Gases or solutes (e.g., oxygen, CO_2)	Water only	glucose, ions, etc.
Membrane needed?	Not always	A partially permeable membrane is required	Requires membrane and carrier proteins

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